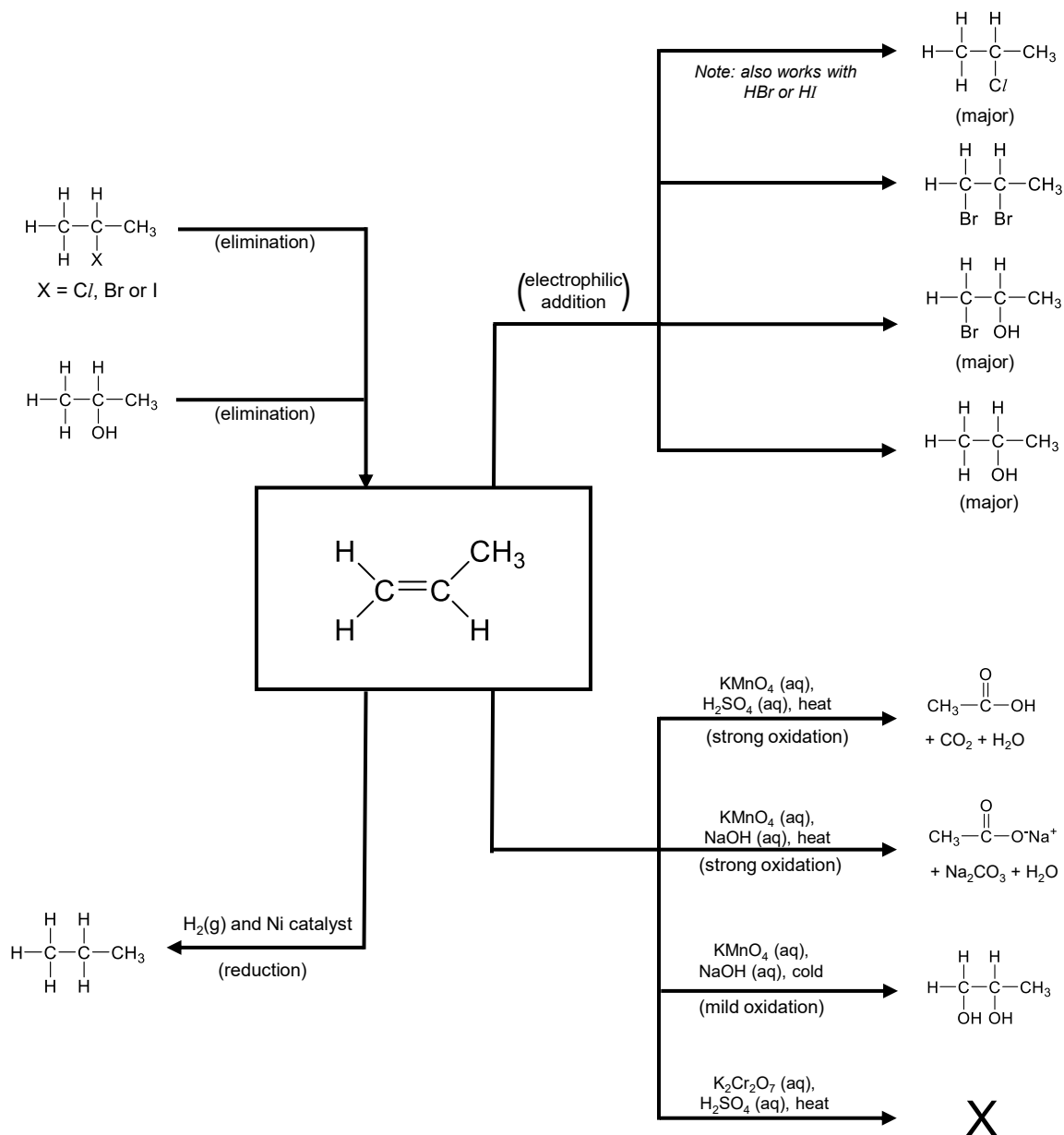
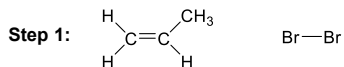


Reactions



Mechanism: with $\text{Br}_2(\text{l})$

Name:



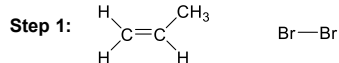
Step 2:

Note:

- product formed is a racemic mixture of 2 enantiomers
- this is because Br^- nucleophile can attack the sp^2 planar positively charged C of the carbocation intermediate from either side of plane, at equal probability, forming a pair of enantiomers with equal proportions

Mechanism: with $\text{Br}_2(\text{aq})$

Name:



Step 2:

Step 3:

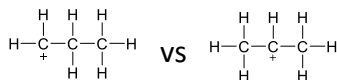
Why electrophilic addition?

- Explain why alkenes are reactive towards electrophilic addition

- 1 Alkenes contain C=C bonds which attract electrophiles
- 2 The bonds in C=C are relatively weak and require less energy to break

Comparing relative stability of carbocation

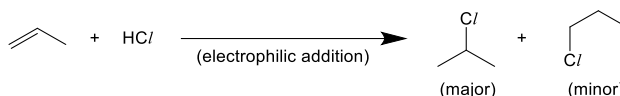
- Circle the more stable carbocation



- Explanation

- in , the positively charged C is bonded to two, as compared to only one in 
- this the positive charge on C to a greater extent, and stabilises the carbocation

Effect of carbocation stability

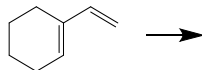
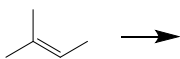


- Explain why 2-chloropropane is formed as the major product whereas 1-chloropropane is formed as the minor product.

- In the first step, and can be formed
- in, the positively charged C is bonded to two electron-donating alkyl group as compared to only one in
- therefore the positive charge is dispersed to a greater extent in
- hence, is a more stable carbocation intermediate. It is formed and is more available to further react to give the major product.

Deducing products from strong oxidation

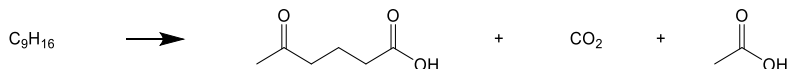
- Deduce the products formed when the following compounds are oxidised using $\text{KMnO}_4(\text{aq})$, $\text{H}_2\text{SO}_4(\text{aq})$, heat



Note: $\text{H}-\text{C}(=\text{O})-\text{OH}$ & $\text{HO}-\text{C}(=\text{O})-\text{C}(=\text{O})-\text{OH}$ are able to further oxidise into $\text{CO}_2 + \text{H}_2\text{O}$; $\text{HO}-\text{C}(=\text{O})-\text{OH}$ can decompose into $\text{CO}_2 + \text{H}_2\text{O}$

Working backwards from strong oxidation

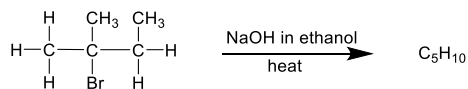
- Deduce possible structures of the alkene which forms the given products upon oxidation with $\text{KMnO}_4(\text{aq})$, $\text{H}_2\text{SO}_4(\text{aq})$, heat
You may ignore stereoisomerism



Deducing products from elimination reaction

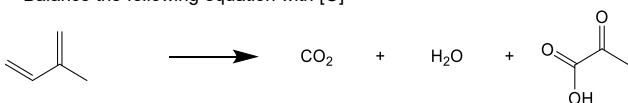
- Draw all possible alkene product formed.
- State the degree of substitution of the alkene formed and hence indicate the major and minor products.

Note: the more substituted alkene is more stable, hence it is the major product



Balancing redox eqⁿ with [H] or [O]

- Balance the following equation with [O]



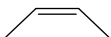
- Balance the following equation with [H]



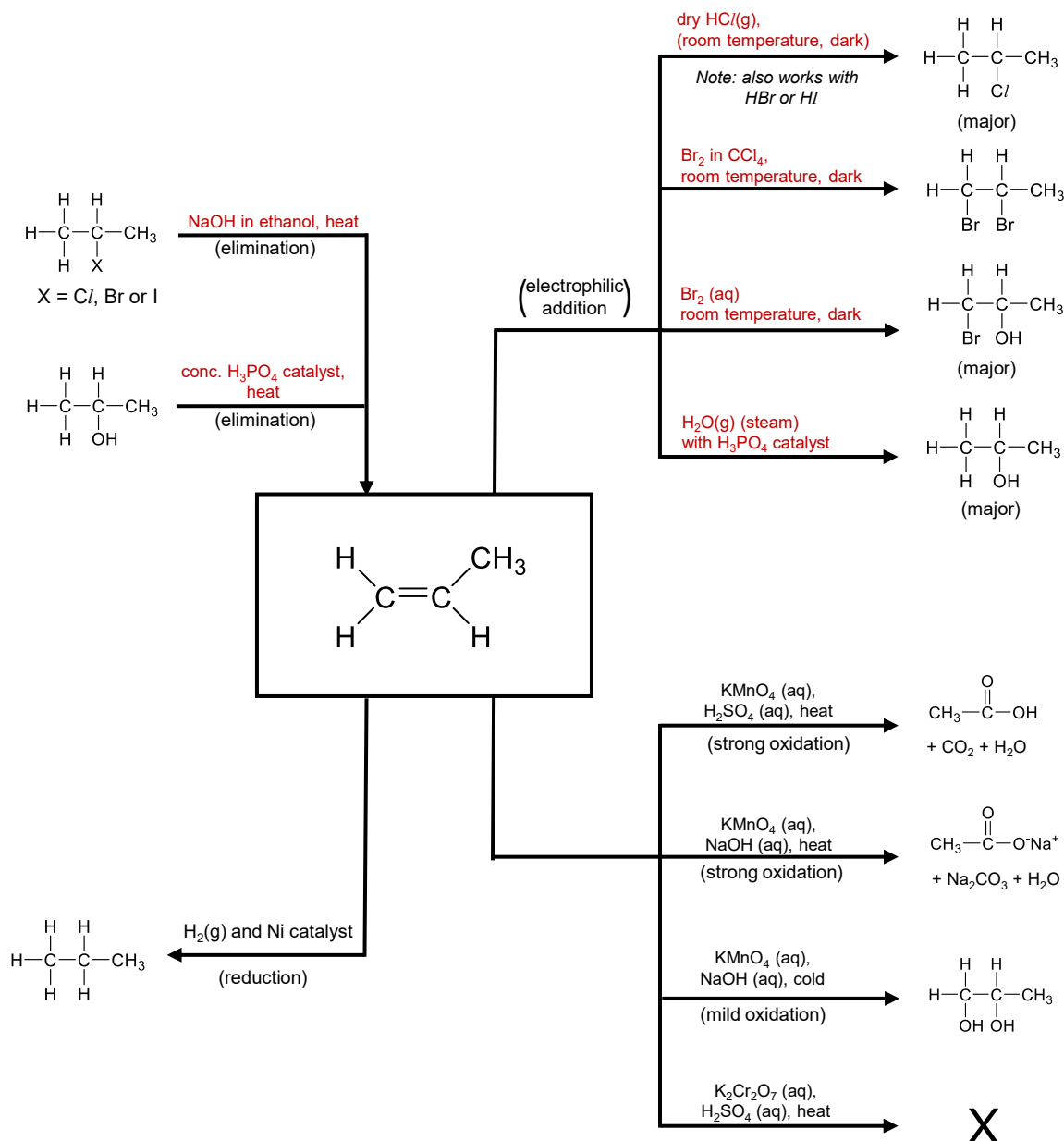
Method

- balance all atoms (other than H / O)
- balance with H₂O
- then balance H with [H] or O with [O]

Physical properties of cis vs trans

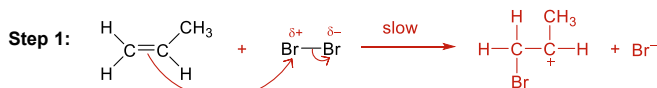
cis vs trans		
melting point	 m.p. • packs efficiently • molecules are further apart • id-id • energy required to overcome the weaker IMF	 m.p. • packs efficiently, • molecules are closer together • id-id • energy required to overcome the stronger IMF
boiling point	 b.p. • polar • has weak pd-pd interactions • energy required to overcome the stronger IMF	 b.p. •- polar • only has id-id interactions • energy required to overcome the weaker IMF

Reactions



Mechanism: with Br₂(l)

Name: electrophilic addition

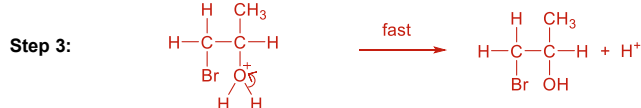
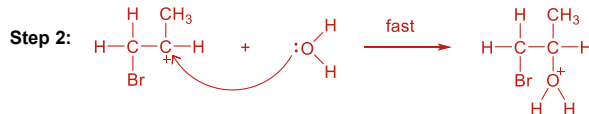
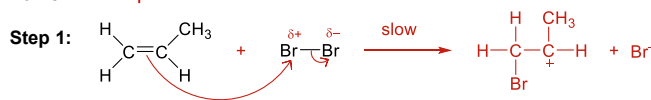


Note:

- product formed is a racemic mixture of 2 enantiomers
- this is because Br⁻ nucleophile can attack the sp² planar positively charged C of the carbocation intermediate from either side of plane, at equal probability, forming a pair of enantiomers with equal proportions

Mechanism: with Br₂(aq)

Name: electrophilic addition



Why electrophilic addition?

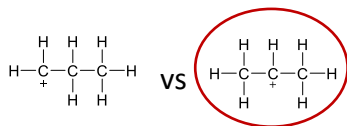
➤ Explain why alkenes are reactive towards electrophilic addition

1 Alkenes contain **electron-rich** C=C bonds which attract electrophiles

2 The **π** bonds in C=C are relatively weak and require less energy to break

Comparing relative stability of carbocation

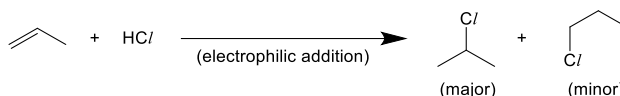
➤ Circle the more stable carbocation



➤ Explanation

- in $\begin{array}{c} \text{H} & \text{H} & \text{H} \\ | & | & | \\ \text{H}-\text{C}^+-\text{C}-\text{C}-\text{H} \\ | & | & | \\ \text{H} & \text{H} & \text{H} \end{array}$, the positively charged C is bonded to two **electron-donating alkyl group** as compared to only one in $\begin{array}{c} \text{H} & \text{H} & \text{H} \\ | & | & | \\ \text{H}-\text{C}-\text{C}^+-\text{C}-\text{H} \\ | & | & | \\ \text{H} & \text{H} & \text{H} \end{array}$
- this **disperses** the positive charge on C to a greater extent, and stabilises the carbocation

Effect of carbocation stability

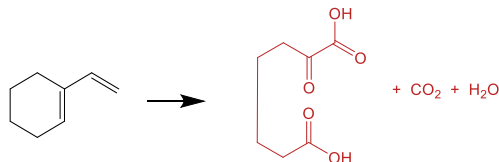


➤ Explain why 2-chloropropane is formed as the major product whereas 1-chloropropane is formed as the minor product.

- In the first step, $\begin{array}{c} \text{H} & \text{H} & \text{H} \\ | & | & | \\ \text{H}-\text{C}^+-\text{C}-\text{C}-\text{H} \\ | & | & | \\ \text{H} & \text{H} & \text{H} \end{array}$ and $\begin{array}{c} \text{H} & \text{H} & \text{H} \\ | & | & | \\ \text{H}-\text{C}-\text{C}^+-\text{C}-\text{H} \\ | & | & | \\ \text{H} & \text{H} & \text{H} \end{array}$ can be formed
- in $\begin{array}{c} \text{H} & \text{H} & \text{H} \\ | & | & | \\ \text{H}-\text{C}^+-\text{C}-\text{C}-\text{H} \\ | & | & | \\ \text{H} & \text{H} & \text{H} \end{array}$, the positively charged C is bonded to two electron-donating alkyl group as compared to only one in $\begin{array}{c} \text{H} & \text{H} & \text{H} \\ | & | & | \\ \text{H}-\text{C}-\text{C}^+-\text{C}-\text{H} \\ | & | & | \\ \text{H} & \text{H} & \text{H} \end{array}$
- therefore the positive charge is dispersed to a greater extent in $\begin{array}{c} \text{H} & \text{H} & \text{H} \\ | & | & | \\ \text{H}-\text{C}^+-\text{C}-\text{C}-\text{H} \\ | & | & | \\ \text{H} & \text{H} & \text{H} \end{array}$
- hence, $\begin{array}{c} \text{H} & \text{H} & \text{H} \\ | & | & | \\ \text{H}-\text{C}^+-\text{C}-\text{C}-\text{H} \\ | & | & | \\ \text{H} & \text{H} & \text{H} \end{array}$ is a more stable carbocation intermediate. It is formed **more rapidly** and is more available to further react to give the major product.

Deducing products from strong oxidation

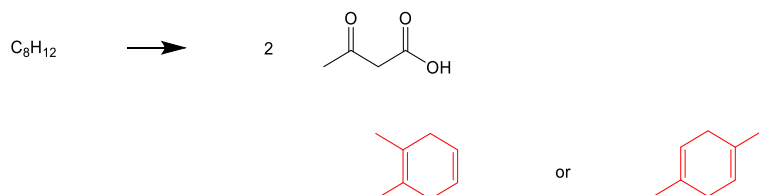
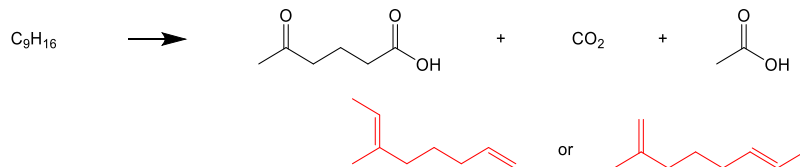
➤ Deduce the products formed when the following compounds are oxidised using $\text{KMnO}_4(\text{aq})$, $\text{H}_2\text{SO}_4(\text{aq})$, heat



Note: $\text{H}-\text{C}(=\text{O})-\text{OH}$ & $\text{HO}-\text{C}(=\text{O})-\text{C}(=\text{O})-\text{OH}$ are able to further oxidise into $\text{CO}_2 + \text{H}_2\text{O}$; $\text{HO}-\text{C}(=\text{O})-\text{OH}$ can decompose into $\text{CO}_2 + \text{H}_2\text{O}$

Working backwards from strong oxidation

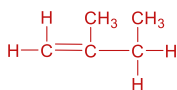
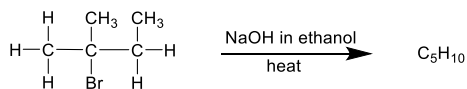
➤ Deduce possible structures of the alkene which forms the given products upon oxidation with $\text{KMnO}_4(\text{aq})$, $\text{H}_2\text{SO}_4(\text{aq})$, heat
You may ignore stereoisomerism



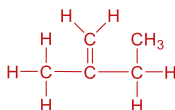
Deducing products from elimination reaction

- Draw all possible alkene product formed.
- State the degree of substitution of the alkene formed and hence indicate the major and minor products.

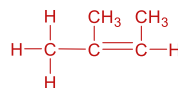
Note: the more substituted alkene is more stable, hence it is the major product



(2°; minor)



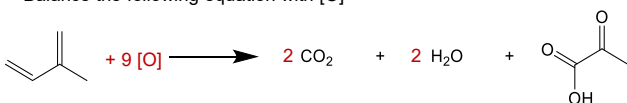
(2°; minor)



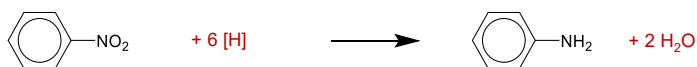
(3°; major)

Balancing redox eqⁿ with [H] or [O]

- Balance the following equation with [O]



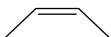
- Balance the following equation with [H]



Method

- balance all atoms (other than H / O)
- balance with H₂O
- then balance H with [H] or O with [O]

Physical properties of cis vs trans

cis vs trans		
melting point	 lower m.p. • packs less efficiently • molecules are further apart • weaker id-id • less energy required to overcome the weaker IMF	 higher m.p. • packs more efficiently, • molecules are closer together • stronger id-id • more energy required to overcome the stronger IMF
boiling point	 higher b.p. • slightly polar • has weak pd-pd interactions • more energy required to overcome the stronger IMF	 lower b.p. • non polar • only has id-id interactions • less energy required to overcome the weaker IMF